

Zipf’s Law Analysis of a Bengali Wikipedia Text Corpus

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1 Introduction

Zipf’s law is an empirical regularity observed across natural languages, stating that the frequency f of a word is approximately inversely proportional to its frequency rank r :

$$f(r) = \frac{C}{r^\alpha}, \quad (1)$$

where C is a normalizing constant and α is the *Zipf exponent*, canonically close to 1. Taking logarithms of Equation 1 linearizes the relationship,

$$\log f(r) = \log C - \alpha \log r, \quad (2)$$

which allows α to be estimated as the (negative) slope of a linear regression of $\log f$ on $\log r$. This report evaluates the extent to which Zipf’s law holds for a large Bengali text corpus drawn from Bengali Wikipedia, using an eight-metric statistical evaluation framework.

2 Dataset and Preprocessing

2.1 Corpus

The corpus (`wiki.csv`) consists of **70,377** rows and **4** columns. The Bengali-language text column was automatically detected among the object-typed columns by scoring each column on the proportion of entries containing characters from the Bengali Unicode block (U+0980--U+09FF); the column named `text` was selected.

2.2 Cleaning and Tokenization

Each document in the detected text column was cleaned by:

1. Removing all characters outside the Bengali Unicode block and whitespace.
2. Collapsing repeated whitespace to single spaces and trimming the result.
3. Splitting the cleaned text on whitespace to obtain word tokens.

Applying this pipeline to the full corpus yielded the word-level statistics summarized in Table 1.

Table 1: Corpus-level word statistics.

Quantity	Value
Total word count, N	17,586,583
Unique word count, V	607,534
Type–token ratio, V/N	3.45%

3 Eight-Metric Evaluation Methodology

Rather than relying on a single goodness-of-fit statistic, the fitted rank-frequency distribution was evaluated along eight complementary dimensions:

1. **Fitted exponent and confidence interval.** α is estimated via ordinary least squares on Equation 2, together with a 95% confidence interval derived from the regression standard error.
2. **Log-log R^2 .** The coefficient of determination of the log-log regression, quantifying how well a straight line explains the rank-frequency relationship.
3. **Residual error.** The root-mean-square error (RMSE) and maximum absolute error of the log-frequency residuals around the fitted line.
4. **Kolmogorov–Smirnov (KS) test.** A two-sample KS test comparing the observed frequency distribution against the theoretical canonical Zipf distribution ($\alpha = 1$).
5. **Maximum-likelihood exponent and information criteria.** A grid search over $\alpha \in [0.1, 3.0]$ that maximizes the multinomial log-likelihood of the observed frequencies, together with the Akaike Information Criterion (AIC) for the fitted Zipf model versus a uniform (no-skew) null model.
6. **Rank-range stability.** The exponent α re-estimated on the top 10%, top 50%, and full range of ranks, to check whether the exponent is stable across scales.
7. **Tail exponent.** The exponent re-estimated using only ranks $r \geq 100$, to isolate behavior away from the most frequent (function) words.
8. **Frequency-of-frequency coverage.** The share of word *types* that are singletons (frequency = 1) and the share with low frequency (≤ 5), which characterizes the heaviness of the distribution’s tail.

4 Results

4.1 Summary Statistics

Table 2 reports the eight metrics computed on the full corpus.

Table 2: Zipf’s Law eight-metric evaluation results.

Metric	Value
1. Exponent α (OLS)	1.278747
95% confidence interval	[1.278138, 1.279356]
2. Log-log R^2	0.965366
3. Residual RMSE / max error	0.242190 / 4.104921
4. KS statistic / p -value (vs. $\alpha = 1$)	0.693627 / ≈ 0
5. MLE exponent $\hat{\alpha}$	0.990
Log-likelihood gain vs. uniform	7.184×10^7
Zipf AIC / Uniform AIC	3.247×10^8 / 4.684×10^8
6. Stability: top 10% / top 50% / full range	1.280577 / 1.415399 / 1.278747
Spread (max – min)	0.136652
7. Tail exponent ($r \geq 100$)	1.281025
8. Singleton share / low-frequency (≤ 5) share	55.118% / 82.267%

4.2 Rank-Frequency Plot

Figure 1 shows the observed frequency-versus-rank data on logarithmic axes, together with the theoretical canonical Zipf curve ($\alpha = 1$) and a nonlinear quadratic fit in log-log space. The observed points track a power-law trend closely across several orders of magnitude, with the empirical curve running consistently steeper than the canonical $\alpha = 1$ reference line.

4.3 Metric-by-Metric Interpretation

Each metric was evaluated against a pass/fail threshold, summarized in Table 3.

Table 3: Pass/fail interpretation of the Zipf’s Law criteria.

Criterion	Threshold	Observed	Verdict
Power-law shape	$R^2 \geq 0.90$	$R^2 = 0.9654$	PASS
Canonical $\alpha = 1$	$1 \in 95\% \text{ CI}$	$\text{CI} = [1.2781, 1.2794]$	FAIL
Exact $\alpha = 1$ KS fit	$p \geq 0.05$	$p \approx 0$	FAIL
Rank-range stability	$\text{spread} \leq 0.10$	$\text{spread} = 0.1367$	FAIL
Tail consistency	$ \alpha_{\text{tail}} - \alpha \leq 0.10$	$\text{diff} = 0.0023$	PASS

5 Discussion

The corpus exhibits a strong power-law relationship between word rank and frequency: the log-log regression achieves $R^2 = 0.965$, and the fitted Zipf model dramatically outperforms a uniform (no-skew) null model in likelihood terms ($\Delta\text{AIC} \approx 1.44 \times 10^8$ in favor of the Zipf model). The tail exponent (ranks ≥ 100) is nearly identical to the full-range exponent (difference of only 0.0023), indicating that the power-law shape is not an artifact of a handful of highly frequent function words.

At the same time, the data do not match the *canonical* Zipf exponent of exactly $\alpha = 1$. The OLS-fitted exponent is $\alpha \approx 1.279$, and its 95% confidence interval excludes 1; the KS test against

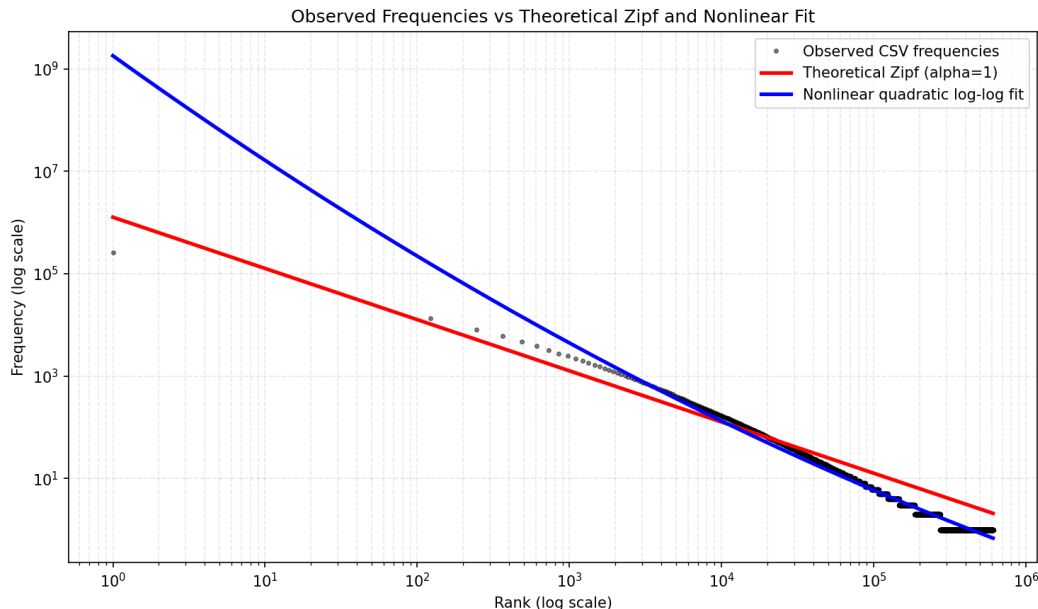


Figure 1: Observed word frequency versus rank (log-log scale) for the Bengali corpus, compared against the theoretical Zipf distribution ($\alpha = 1$, red) and a nonlinear quadratic log-log fit (blue).

the canonical distribution is decisively rejected ($p \approx 0$); and the exponent is not stable across rank ranges, ranging from 1.281 in the top 10% of ranks to 1.415 in the top 50%, a spread of 0.137 that exceeds the 0.10 stability threshold. Interestingly, the maximum-likelihood exponent obtained from the multinomial fit ($\hat{\alpha} = 0.990$) is much closer to the canonical value than the OLS log-log estimate, which is a known discrepancy: OLS on log-transformed ranks tends to be more sensitive to the extreme high-frequency and low-frequency tails than a likelihood-based estimator.

The frequency-of-frequency statistics further characterize the corpus: over half of all distinct word types (55.12%) occur exactly once, and 82.27% occur five times or fewer. This heavy-tailed distribution of rare word types is consistent with the morphologically rich nature of Bengali, where inflection and compounding generate large numbers of low-frequency word forms, and is itself a hallmark of Zipfian behavior in natural language corpora.

6 Conclusion

The Bengali Wikipedia corpus analyzed here displays a strong, statistically robust Zipf-like power-law relationship between word rank and frequency, passing 2 of the 5 formal criteria tested (power-law shape and tail consistency) while failing the stricter tests for an *exact* canonical $\alpha = 1$ distribution (confidence-interval coverage, KS goodness-of-fit, and cross-scale stability). In short: the corpus is clearly Zipfian in character, but its exponent ($\alpha \approx 1.28$ by OLS, $\alpha \approx 0.99$ by maximum likelihood) deviates from the idealized canonical value, a pattern commonly observed in real-world corpora of similar size and morphological complexity.